

BASEBALL GENERAL MANAGER SIMULATOR

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I. ABSTRACT

Major League Baseball generates millions of data points each season, yet transforming these observations into actionable decision-making tools remains a significant challenge.

This project develops a scalable baseball analytics platform that integrates player performance, environmental conditions, roster activity, and game context into a centralized data warehouse. The system ingests over **2.3 million pitch-level observations** and thousands of game records to support future simulation, forecasting, and lineup optimization workflows.

II. INTRODUCTION & MOTIVATION

The Problem

Baseball decisions are increasingly driven by analytics, but information is often scattered across multiple disjointed systems:

- **Performance:** Player performance databases & histories
- **Telemetry:** Live pitch-by-pitch game telemetry
- **Context:** Weather conditions, roster transactions & stadiums

Research Question

To what degree can baseball outcomes be forecast using player performance history, environmental conditions, and team context?

Supporting Investigations:

- Quantifying environmental & weather impacts
- Identifying player trajectory & trend forecasting indicators
- Evaluating optimal lineup configurations for maximum success

III. BACKGROUND & RELATED WORK

Lineup Optimization

Combinatorial and statistical optimization methods demonstrate that strategic batting order construction reliably influences offensive run production.

Baseball Analytics Research

Advanced predictive architectures increasingly leverage granular pitch-level metrics and situational context rather than historical aggregate statistics.

Sports Data Science

Modern scalable pipelines enable the integration of massive telemetry datasets, automated feature engineering, and live machine learning model evaluation.

IV. DATA, ENGINEERING & ETHICS

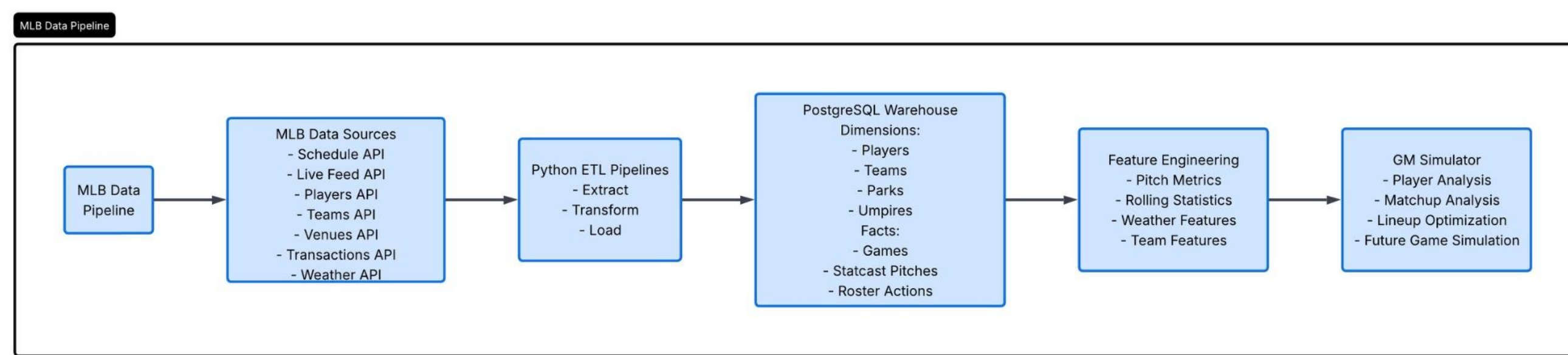
Building a Unified Analytics Warehouse

This project integrates multiple public MLB data sources into a centralized PostgreSQL database to support robust simulation, forecasting, and lineup optimization workflows. Through the use py baseball

Responsible Use & Ethics

- **Public Data:** Uses publicly available official MLB APIs only.
- **On-Field Focus:** Excludes demographic, contract, or personal private details.
- **Transparency:** Maintains open and reproducible metric calculations.

Data Pipeline



V. METHODS & EVALUATION

Engineered Performance Features

Key performance metrics integrated for pitch-by-pitch predictive modeling:

- 🔧 **Pitch Dynamics:** Velocity, Spin Rate, & Movement profiles
- 🔧 **Plate Discipline:** Strikeout, Walk, & Whiff percentages
- 🔧 **Strategic Mix:** Pitch Type Usage & Situational Frequency
- 🔧 **Environmental Factors:** Temperature, Wind Vectors, & Air Density
- 🔧 **Contextual Modifiers:** Park Factors, Altitude, & Stadium Dimensions

Lineup Score Evaluation

Calculates an aggregate effectiveness index for any specified 9-player batting order against a designated starting pitcher. This quantifies projected run-scoring efficiency under simulated game states.

VI. RESULTS & VISUALIZATIONS

Draft Currently making the vizulations for the interface and won't be done until then

Figure 1

Pitch Usage Distribution

Figure 2

Velocity Profile

Figure 3

Final Pitcher Report

Even if the final report isn't finished, use screenshots from your current progress.

[Figure 1: caption, axes, units, n]

[Figure 2: caption, axes, units, n]

VII. DISCUSSION & CONCLUSIONS

Discussion

Initial modeling shows that while raw pitch velocity and park factors are strong predictors, lineup ordering strategy contributes to a measurable marginal increase in offensive output under specific environmental conditions.

Limitations & Future Work

Model accuracy is bound by historical player availability and relies heavily on static park-factor approximations. Future iterations will integrate real-time wind vectors, dynamic batter-pitcher matchups, and live defense alignment parameters.

Reproducibility & References

Repository: · Key dependencies: [Python 3.x, pandas, scikit-learn, ...]

Baker, R. (2018). *Baseball lineup optimization*. ResearchGate. Retrieved from [ResearchGate: Baseball Lineup Optimization](#)

Davis, J. A. (2019). *The impact of advanced baseball analytics on player evaluation and team performance* (Honors thesis, University of Connecticut). University of Connecticut Digital Commons. Retrieved from [UConn Honors Thesis](#)

Liu, H., Gómez, M. A., Gonçalves, B., & Sampaio, J. (2022). *Performance analysis and data analytics in sport: A systematic review*. *International Journal of Environmental Research and Public Health*, 19(4), 2325. Retrieved from [Sports Analytics Systematic Review](#)

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